

SAINT-MICHEL-DES-SAINTS, QC, Canada

WMO: 717370

Lat: 46.816N

Lon: 74.094W

Elev: 1410

StdP: 13.96

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-16.9	-28.1	1.2	-21.0	-23.0	1.6	-15.3	16.4	11.4	14.4	10.1	1.4	280	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	82.2	66.0	79.3	65.1	76.5	63.5	69.8	77.4	68.1	74.4	66.5	73.1	7.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	106.3	73.0	65.9	100.5	71.5	64.0	94.1	70.1	34.8	77.7	33.3	74.2	32.0	73.1	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.1	10.7	-29.5	87.3	3.5	2.4	-32.0	89.0	-34.1	90.5	-36.1	91.8	-38.7	93.5		
(5)			-30.7	72.8	5.5	1.3	-34.7	73.7	-37.9	74.5	-41.0	75.2	-45.1	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	8.8	10.8	21.7	34.8	50.6	57.9	63.8	61.8	54.3	41.8	28.9	15.4		
	DBStd	21.73	13.50	12.33	12.31	9.34	8.67	6.53	5.32	5.28	8.14	8.21	10.23	12.75		
	HDD50	5851	1278	1098	879	462	99	7	0	0	46	275	635	1072		
	HDD65	10078	1743	1518	1343	907	452	231	86	127	331	719	1084	1537		
	CDD50	1357	0	0	1	5	118	243	429	364	176	21	0	0		
	CDD65	112	0	0	0	0	7	17	50	27	11	0	0	0		
	CDH74	1190	0	0	0	1	132	202	543	220	91	1	0	0		
	CDH80	189	0	0	0	0	26	28	99	23	13	0	0	0		
	WSAvg	4.4	4.8	5.0	5.2	5.5	4.6	4.3	3.9	3.6	3.6	4.3	4.4	4.2		
	PrecAvg	40.5	3.1	2.3	2.5	2.8	3.3	4.0	4.2	4.1	4.1	3.7	3.2	3.2		
(15) Precipitation	PrecMax	48.6	5.9	5.8	4.8	5.9	5.2	6.7	6.9	7.5	9.0	5.7	5.9	5.1		
	PrecMin	34.3	1.3	1.0	0.4	0.6	1.8	1.7	2.5	1.9	1.7	1.0	1.0	1.4		
	PrecStd	3.3	1.1	1.0	1.1	1.1	1.1	1.3	1.2	1.3	1.7	1.3	1.1	1.1		
	PrecStd	3.3	1.1	1.0	1.1	1.1	1.1	1.3	1.2	1.3	1.7	1.3	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	42.9	61.1	67.3	83.2	83.4	87.1	82.6	82.0	70.6	57.0	43.2		
		MCWB	45.0	37.7	53.3	55.7	63.0	65.1	69.8	66.8	69.1	58.4	50.8	41.8		
	2%	DB	38.0	35.7	48.3	60.0	77.5	79.7	82.8	79.4	76.3	64.1	51.1	38.7		
		MCWB	35.8	33.2	39.8	47.8	59.4	63.2	67.1	66.5	65.9	55.2	44.7	37.5		
	5%	DB	32.6	32.4	43.3	54.8	72.9	75.9	79.7	76.2	72.5	60.0	46.5	34.5		
		MCWB	31.4	30.3	35.9	44.9	57.1	61.6	65.9	64.5	63.7	53.5	42.3	33.2		
	10%	DB	28.4	29.9	39.0	49.8	67.7	72.2	76.8	73.3	69.1	55.5	42.4	32.3		
		MCWB	27.1	28.3	33.4	41.1	54.5	59.2	64.5	63.8	62.4	51.1	39.5	31.2		
	0.4%	WB	45.2	41.4	54.1	57.5	67.0	70.2	72.6	70.7	70.3	61.1	52.9	42.1		
		MCDB	45.8	42.0	61.2	66.1	76.7	79.0	81.5	77.8	77.8	66.3	55.4	42.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.3	33.7	42.1	51.6	64.1	67.2	70.5	68.4	67.7	57.7	46.4	37.6		
		MCDB	37.3	35.3	45.6	55.9	72.7	74.2	78.2	74.9	72.8	62.4	49.9	38.6		
	5%	WB	31.5	31.1	37.3	46.1	60.9	64.8	68.5	66.8	65.8	54.6	42.5	33.4		
		MCDB	32.7	32.5	42.5	52.1	67.3	70.9	75.0	73.2	70.8	59.0	45.4	34.2		
	10%	WB	27.2	28.7	34.2	42.2	57.6	62.2	66.7	65.1	63.4	51.3	39.7	31.2		
		MCDB	28.3	30.3	38.1	49.1	63.6	69.2	73.9	71.3	68.3	54.4	42.0	32.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	20.7	21.2	19.6	22.7	21.5	21.8	20.1	20.6	16.6	14.9	15.7		
		MCDBR	18.7	18.2	26.2	29.0	31.5	27.1	26.1	24.6	24.0	25.6	20.7	14.3		
		MCWBR	18.1	16.3	17.8	17.9	15.3	13.0	12.9	13.1	14.3	17.3	15.8	14.0		
		MCDBR	18.9	18.3	22.1	24.2	23.8	19.9	21.8	20.1	21.1	22.5	17.7	14.5		
	5% WB	MCWBR	18.5	17.1	16.0	17.4	13.7	11.8	12.7	11.6	14.1	17.0	15.0	14.4		
		MCWBR	18.5	17.1	16.0	17.4	13.7	11.8	12.7	11.6	14.1	17.0	15.0	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.269	0.298	0.345	0.356	0.381	0.381	0.372	0.352	0.321	0.291	0.257		
	taud		2.315	2.256	2.253	2.336	2.419	2.350	2.361	2.390	2.439	2.530	2.501	2.361		
	Ebn at Noon		270	287	294	288	288	280	279	277	273	267	255	254		
	Edh at Noon		23	31	36	37	36	38	37	35	30	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		424	725	1161	1459	1617	1761	1717	1506	1170	676	392	299		
	RadStd		66	102	111	124	160	128	168	77	103	66	53	26		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page